

数学作业纸

科目 _____

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$$\begin{aligned} 3.8 \quad e_{ss} &= \lim_{s \rightarrow 0} s \cdot \frac{R(s)}{1+G(s)H(s)} \\ &= \lim_{s \rightarrow 0} s \cdot \frac{\frac{1}{s^2}}{1 + \frac{K(0.5s+1)}{s(s+1)(2s+1)}} \\ &= \lim_{s \rightarrow 0} \frac{(s+1)(2s+1)}{s(s+1)(2s+1) + K(0.5s+1)} \\ &= \frac{1}{K} < \frac{1}{5} \end{aligned}$$

当 $K > 5$ 时, 稳态误差小于 0.2

$$\begin{aligned} 3.9 \quad \frac{X_c}{X_r} &= \frac{\frac{10}{s} \cdot \frac{10}{s(s+1)}}{1 + \frac{10}{s} \cdot \frac{10}{s(s+1)}} \\ &= \frac{10(2s+1)}{s^2(s+1) + 10(2s+1)} \\ &= \frac{10(2s+1)}{s^3 + s^2 + 10 \cdot 2s + 10} \end{aligned}$$

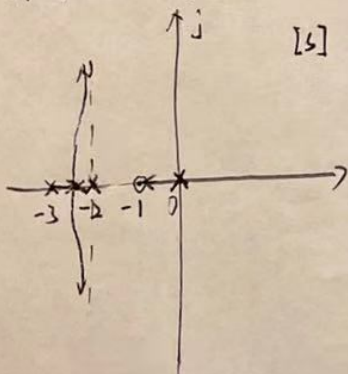
s^3	1	10
s^2	1	10
s^1	$10(2-1)$	0
s^0	10	

当 $10(2-1) > 0$

即 $2 > 1$ 时, 系统稳定

4.3 开环零点 (-1) , $m=1$

开环极点 $(0, -2, -3)$, $n=3$



始于 $0, -2, -3$, 终于 $-1, \infty$, 实轴 $[-3, -2]$, $[-1, 0]$ 有轨迹

系统闭环特征方程:

$$s^4 + 4s^3 + 6s^2 + 4s + K^* = 0$$

s^4	1	8	K^*
s^3	4	4	0
s^2	5	K^*	0
s^1	$\frac{1(20-4K^*)}{5}$	0	0
s^0	K^*	0	0

$$\therefore 20 - 4K^* = 0, K^* = 5$$

解辅助方程:

$$5s^2 + K^* = 0, s = \pm j$$

$$\sigma = \frac{0-2-3-(-1)}{3-1} = -2$$

$$\phi = \frac{(2k+1)\pi}{3-1} \quad k=0, 1 \quad \alpha = \frac{2k+1}{1}\pi$$

$$\phi = \pm \frac{\pi}{2} \quad = \mp \frac{\pi}{2}$$

$\therefore$ 渐近线垂直于 X 轴, 且交点为 -2

$$\frac{1}{d+3} + \frac{1}{d+2} + \frac{1}{d} = \frac{1}{d+1}$$

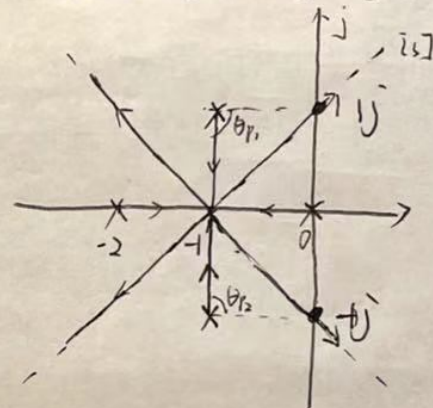
$$d \approx -2.47$$

分离点坐标 $(-2.47, 0)$

分离角 $\pm \frac{\pi}{2}$

4.4 开环零点 $m=0$

开环极点 $(0, -2, j-1, -j-1)$ $n=4$



始于 $0, -2, j-1, -j-1$, 终于 ∞

实轴 $[-2, 0]$ 有轨迹

$$\sigma = \frac{0-2+j-1-j-1}{4-0} = -1$$

$$\phi = \frac{(2k+1)\pi}{4-0} = \frac{2k+1}{4}\pi \quad k=0, 1, 2, 3$$

$$\phi = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4} \quad \alpha = \phi$$

$$\frac{1}{d+2} + \frac{1}{d} + \frac{1}{d-(j-1)} + \frac{1}{d-(-j-1)} = 0$$

$$|c| = -1 \quad \text{分离角: } \frac{\pi}{4}, \frac{3\pi}{4}, -\frac{\pi}{4}, -\frac{3\pi}{4}$$

$$\theta_{p1} = 180^\circ - 135^\circ - 90^\circ - 45^\circ = -90^\circ$$

由对称性 $\theta_{p2} = 90^\circ$

$$s(s+2)(s^2+2s+2) + K^* = 0$$

$$j\omega(j\omega+2)[(j\omega)^2+2j\omega+2] + K^* = 0$$

$$(4\omega - 4\omega^3)j + (\omega^4 - 6\omega^2 + K^*) = 0$$

$$\begin{cases} \omega=1 \\ \omega=2 \end{cases} \quad \begin{cases} \omega=1 \\ \omega=2 \end{cases} \quad \begin{cases} \omega=1 \\ \omega=2 \end{cases}$$

```
num=[1];  
den = [1 4 6 4 0];  
G = tf(num,den);  
rlocus(G)
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